Customer
Analytics Dashboard - Master Prompt v5 (Locked UI Version)

Here is the complete, self-contained VBA module written according to your exact specifications, including all requested Pivot Tables, KPI formulas, Named Ranges, KPI Cards, Charts, Slicers, and the required helper procedures.

VBA

```
Option Explicit

Sub GenerateCustomerAnalyticsDashboard()
    Dim wb As Workbook
    Dim wsData As Worksheet, wsPivot As Worksheet, wsDash As Worksheet
    Dim rngData As Range
    Dim loTable As ListObject
    Dim pc As PivotCache
    Dim ptTrend As PivotTable, ptAgent As PivotTable, ptRegion As PivotTable
    Dim ptProduct As PivotTable, ptConversion As PivotTable
    Dim ptTable As PivotTable
    Dim pfVal As PivotField
    Dim choTrend As ChartObject, choAgent As ChartObject, choRegion As ChartObj
    Dim choProduct As ChartObject, choConversion As ChartObject
    Dim scYear As SlicerCache, scCustType As SlicerCache, scProduct As SlicerCa
    Dim slYear As Slicer, slCustType As Slicer, slProduct As Slicer
    Dim shpHeader As Shape
    Dim currentLeft As Double

    Set wb = ActiveWorkbook

    '---------------------------------------------------------------------------
    ' 0. PREPARATION & CLEANUP (Idempotent)
    '---------------------------------------------------------------------------
    Application.ScreenUpdating = False
    Application.DisplayAlerts = False

    On Error Resume Next
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    wb.Worksheets("Dashboard").Delete
    wb.Worksheets("Pivot").Delete
    On Error GoTo 0

    Application.DisplayAlerts = True

    ' Verify Data Sheet
    On Error Resume Next
    Set wsData = wb.Worksheets("Customer_Data")
    On Error GoTo 0
    If wsData Is Nothing Then
        MsgBox "Sheet 'Customer_Data' not found! Please check worksheet name.",
        Application.ScreenUpdating = True
        Exit Sub
    End If

    ' Format Data Range as Table
    Set rngData = wsData.Range("A1").CurrentRegion
    On Error Resume Next
    Set loTable = wsData.ListObjects("tbl_CustomerData")
    On Error GoTo 0
    If loTable Is Nothing Then
        Set loTable = wsData.ListObjects.Add(xlSrcRange, rngData, , xlYes)
        loTable.Name = "tbl_CustomerData"
    End If

    ' Create Sheets
    Set wsPivot = wb.Worksheets.Add(After:=wb.Worksheets(wb.Worksheets.Count))
    wsPivot.Name = "Pivot"

    Set wsDash = wb.Worksheets.Add(Before:=wb.Worksheets(1))
    wsDash.Name = "Dashboard"

    ' Create Pivot Cache
    Set pc = wb.PivotCaches.Create(SourceType:=xlDatabase, SourceData:="tbl_Cus

    '--------------------------------------------------------------------------
    ' 1. BUILD PIVOT TABLES
    '--------------------------------------------------------------------------

    ' PT_Trend (Line) - Pos: F3
    Set ptTrend = pc.CreatePivotTable(TableDestination:=wsPivot.Range("F3"), Ta
    With ptTrend
        .PivotFields("Month").Orientation = xlRowField
        .PivotFields("Year").Orientation = xlColumnField
        Set pfVal = .AddDataField(.PivotFields("Revenue Generated"), "Sum of Re
        pfVal.NumberFormat = "$#,##0"
    End With
```

```
    ' PT_Agent (Column) - Pos: Q3
    Set ptAgent = pc.CreatePivotTable(TableDestination:=wsPivot.Range("Q3"), Ta
    With ptAgent
        .PivotFields("Agent Name").Orientation = xlRowField
        Set pfVal = .AddDataField(.PivotFields("Revenue Generated"), "Sum of Re
        pfVal.NumberFormat = "$#,##0"
        .PivotFields("Agent Name").AutoSort xlDescending, pfVal.Name
    End With

    ' PT_Region (Column) - Pos: K3
    Set ptRegion = pc.CreatePivotTable(TableDestination:=wsPivot.Range("K3"), T
    With ptRegion
        .PivotFields("Region").Orientation = xlRowField
        Set pfVal = .AddDataField(.PivotFields("Revenue Generated"), "Sum of Re
        pfVal.NumberFormat = "$#,##0"
        .PivotFields("Region").AutoSort xlDescending, pfVal.Name
    End With

    ' PT_Product (Bar) - Pos: N3
    Set ptProduct = pc.CreatePivotTable(TableDestination:=wsPivot.Range("N3"),
    With ptProduct
        .PivotFields("Product/Service").Orientation = xlRowField
        Set pfVal = .AddDataField(.PivotFields("Revenue Generated"), "Sum of Re
        pfVal.NumberFormat = "$#,##0"
        .PivotFields("Product/Service").AutoSort xlDescending, pfVal.Name
    End With

    ' PT_Conversion (Doughnut) - Pos: T3
    Set ptConversion = pc.CreatePivotTable(TableDestination:=wsPivot.Range("T3"
    With ptConversion
        .PivotFields("Conversion Status").Orientation = xlRowField
        .AddDataField .PivotFields("Lead ID"), "Count of Lead ID", xlCount
    End With

    '--------------------------------------------------------------------------
    ' 2. PT_KPI HELPER RANGE (C3:D8) & NAMED RANGES
    '--------------------------------------------------------------------------
    wsPivot.Range("C3").Value = "Total Revenue"
    wsPivot.Range("D3").Formula = "=SUM(tbl_CustomerData[Revenue Generated])"
    wsPivot.Range("D3").NumberFormat = "$#,##0"
    wb.Names.Add Name:="KPI_TotalRevenue", RefersTo:="=Pivot!$D$3"

    wsPivot.Range("C4").Value = "Total Leads"
    wsPivot.Range("D4").Formula = "=COUNTA(tbl_CustomerData[Lead ID])"
    wsPivot.Range("D4").NumberFormat = "#,##0"
    wb.Names.Add Name:="KPI_TotalLeads", RefersTo:="=Pivot!$D$4"

    wsPivot.Range("C5").Value = "Avg Rev / Lead"
    wsPivot.Range("D5").Formula = "=KPI_TotalRevenue/KPI_TotalLeads"
```

```
    wsPivot.Range("D5").NumberFormat = "$#,##0.00"
    wb.Names.Add Name:="KPI_AvgRevLead", RefersTo:="=Pivot!$D$5"

    wsPivot.Range("C6").Value = "Converted Leads"
    wsPivot.Range("D6").Formula = "=COUNTIF(tbl_CustomerData[Conversion Status]
    wsPivot.Range("D6").NumberFormat = "#,##0"
    wb.Names.Add Name:="KPI_ConvertedLeads", RefersTo:="=Pivot!$D$6"

    wsPivot.Range("C7").Value = "Conversion Rate"
    wsPivot.Range("D7").Formula = "=KPI_ConvertedLeads/KPI_TotalLeads"
    wsPivot.Range("D7").NumberFormat = "0.0%"
    wb.Names.Add Name:="KPI_ConversionRate", RefersTo:="=Pivot!$D$7"

    wsPivot.Range("C8").Value = "Unique Agents"
    wsPivot.Range("D8").Formula = "=SUMPRODUCT(1/COUNTIF(tbl_CustomerData[Agent
    wsPivot.Range("D8").NumberFormat = "#,##0"
    wb.Names.Add Name:="KPI_UniqueAgents", RefersTo:="=Pivot!$D$8"

    '--------------------------------------------------------------------------
    ' 3. DASHBOARD UI SETUP
    '--------------------------------------------------------------------------
    ' Hide Gridlines
    wsDash.Activate
    ActiveWindow.DisplayGridlines = False

    ' Header Banner (A1:Y4)
    Set shpHeader = wsDash.Shapes.AddShape(msoShapeRectangle, _
        wsDash.Range("A1").Left, wsDash.Range("A1").Top, _
        wsDash.Range("A1:Y4").Width, wsDash.Range("A1:Y4").Height)

    With shpHeader
        .Fill.ForeColor.RGB = RGB(0, 112, 192)
        .Line.Visible = msoFalse
        With .TextFrame2.TextRange
            .Text = "CUSTOMER ANALYTICS DASHBOARD"
            .Font.Name = "Segoe UI"
            .Font.Size = 28
            .Font.Bold = msoTrue
            .Font.Fill.ForeColor.RGB = RGB(255, 255, 255)
            .ParagraphFormat.Alignment = msoAlignCenter
        End With
    End With

    '--------------------------------------------------------------------------
    ' 4. DRAW KPI CARDS
    '--------------------------------------------------------------------------
    currentLeft = 15

    Call DrawKPICard(wsDash, "Total Revenue", "KPI_TotalRevenue", RGB(91, 155,
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    currentLeft = currentLeft + 204

    Call DrawKPICard(wsDash, "Total Leads", "KPI_TotalLeads", RGB(255, 192, 0),
    currentLeft = currentLeft + 204

    Call DrawKPICard(wsDash, "Avg Rev / Lead", "KPI_AvgRevLead", RGB(162, 74, 2
    currentLeft = currentLeft + 204

    Call DrawKPICard(wsDash, "Converted Leads", "KPI_ConvertedLeads", RGB(237,
    currentLeft = currentLeft + 204

    Call DrawKPICard(wsDash, "Conversion Rate", "KPI_ConversionRate", RGB(20, 1
    currentLeft = currentLeft + 204

    Call DrawKPICard(wsDash, "Unique Agents", "KPI_UniqueAgents", RGB(0, 176, 2

    '-------------------------------------------------------------------------
    ' 5. CREATE & STYLE CHARTS
    '-------------------------------------------------------------------------

    ' Chart 1: Trend
    Set choTrend = wsDash.ChartObjects.Add(Left:=130, Top:=155, Width:=500, Hei
    choTrend.Chart.SetSourceData ptTrend.TableRange1
    Call ApplyProfessionalStyling(choTrend.Chart, "Trend")

    ' Chart 2: Agent
    Set choAgent = wsDash.ChartObjects.Add(Left:=640, Top:=155, Width:=555, Hei
    choAgent.Chart.SetSourceData ptAgent.TableRange1
    Call ApplyProfessionalStyling(choAgent.Chart, "Agent")

    ' Chart 3: Region
    Set choRegion = wsDash.ChartObjects.Add(Left:=130, Top:=435, Width:=330, He
    choRegion.Chart.SetSourceData ptRegion.TableRange1
    Call ApplyProfessionalStyling(choRegion.Chart, "Region")

    ' Chart 4: Product
    Set choProduct = wsDash.ChartObjects.Add(Left:=470, Top:=435, Width:=400, H
    choProduct.Chart.SetSourceData ptProduct.TableRange1
    Call ApplyProfessionalStyling(choProduct.Chart, "Product")

    ' Chart 5: Conversion
    Set choConversion = wsDash.ChartObjects.Add(Left:=880, Top:=435, Width:=315
    choConversion.Chart.SetSourceData ptConversion.TableRange1
    Call ApplyProfessionalStyling(choConversion.Chart, "Conversion")

    '-------------------------------------------------------------------------
    ' 6. SLICERS
    '-------------------------------------------------------------------------
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    ' Year Slicer
    Set scYear = wb.SlicerCaches.Add2(ptTrend, "Year")
    Set slYear = scYear.Slicers.Add(wsDash, , "Slicer_Year", "Year", 155, 15, 1
    slYear.Style = "SlicerStyleDark1"

    ' Customer Type Slicer
    Set scCustType = wb.SlicerCaches.Add2(ptTrend, "Customer Type")
    Set slCustType = scCustType.Slicers.Add(wsDash, , "Slicer_CustomerType", "C
    slCustType.Style = "SlicerStyleDark1"

    ' Product/Service Slicer
    Set scProduct = wb.SlicerCaches.Add2(ptTrend, "Product/Service")
    Set slProduct = scProduct.Slicers.Add(wsDash, , "Slicer_Product", "Product/
    slProduct.Style = "SlicerStyleDark1"

    ' Connect Slicers to other Data Pivot Tables (ptAgent, ptRegion, ptProduct,
    scYear.PivotTables.AddPivotTable ptAgent
    scYear.PivotTables.AddPivotTable ptRegion
    scYear.PivotTables.AddPivotTable ptProduct
    scYear.PivotTables.AddPivotTable ptConversion

    scCustType.PivotTables.AddPivotTable ptAgent
    scCustType.PivotTables.AddPivotTable ptRegion
    scCustType.PivotTables.AddPivotTable ptProduct
    scCustType.PivotTables.AddPivotTable ptConversion

    scProduct.PivotTables.AddPivotTable ptAgent
    scProduct.PivotTables.AddPivotTable ptRegion
    scProduct.PivotTables.AddPivotTable ptProduct
    scProduct.PivotTables.AddPivotTable ptConversion

    Application.ScreenUpdating = True
End Sub


'==============================================================================
' UI HELPER FUNCTIONS
'==============================================================================

Public Sub DrawKPICard(ws As Worksheet, cardTitle As String, cellFormulaLink As
    Dim shBody As Shape, shTitle As Shape, shVal As Shape
    Set shBody = ws.Shapes.AddShape(msoShapeRoundedRectangle, L, T, W, H)
    With shBody
        .Fill.ForeColor.RGB = accentClr
        .Line.Visible = msoFalse
        .Shadow.Visible = msoFalse
    End With
    Set shTitle = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, L + 6, T +
    With shTitle
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        .Name = "KPI_Title_" & safeKey
        .Fill.Visible = msoFalse
        .Line.Visible = msoFalse
        With .TextFrame2.TextRange
            .Text = UCase(cardTitle)
            .Font.Name = "Segoe UI": .Font.Size = 10: .Font.Bold = msoTrue
            .Font.Fill.ForeColor.RGB = RGB(255, 255, 255)
            .ParagraphFormat.Alignment = msoAlignCenter
        End With
    End With
    Set shVal = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, L + 4, T + 2
    With shVal
        .Name = "KPI_Val_" & safeKey
        .Fill.Visible = msoFalse
        .Line.Visible = msoFalse
        .DrawingObject.Formula = "=" & cellFormulaLink
        With .TextFrame2.TextRange
            .Font.Name = "Segoe UI": .Font.Size = 20: .Font.Bold = msoTrue
            .Font.Fill.ForeColor.RGB = RGB(255, 255, 255)
            .ParagraphFormat.Alignment = msoAlignCenter
        End With
    End With
End Sub

Public Sub ApplyProfessionalStyling(ch As Chart, chartType As String)
    Dim s As Series, i As Long, pName As String
    On Error Resume Next
    ch.ShowAllFieldButtons = False
    ch.HasTitle = True
    ch.ChartTitle.Font.Name = "Segoe UI"
    ch.ChartTitle.Font.Bold = True
    ch.ChartTitle.Font.Size = 12
    ch.Axes(xlValue).HasMajorGridlines = False

    Select Case chartType
        Case "Trend"
            ch.ChartType = xlLineMarkersStacked
            ch.ChartTitle.Text = "Revenue Trend by Month"
            ch.HasLegend = False
            ch.SetElement msoElementDataLabelTop
            ch.Axes(xlCategory).AxisBetweenCategories = False
            If ch.SeriesCollection.Count >= 1 Then
                ch.SeriesCollection(1).Format.Line.ForeColor.RGB = RGB(0, 176,
                ch.SeriesCollection(1).Smooth = True
            End If
            If ch.SeriesCollection.Count >= 2 Then
                ch.SeriesCollection(2).Format.Line.ForeColor.RGB = RGB(255, 0,
                ch.SeriesCollection(2).Smooth = True
            End If
```

```
        Case "Agent"
            ch.ChartType = xlColumnClustered
            ch.ChartGroups(1).GapWidth = 20
            ch.ChartTitle.Text = "Top Performing Agents"
            ch.HasLegend = False
            ch.SeriesCollection(1).Format.Fill.ForeColor.RGB = RGB(237, 125, 49
            ch.SeriesCollection(1).Format.Line.Visible = msoTrue
            ch.SeriesCollection(1).Format.Line.ForeColor.RGB = RGB(127, 127, 12
        Case "Region"
            ch.ChartType = xlColumnClustered
            ch.ChartTitle.Text = "Revenue by Region"
            ch.HasLegend = False
            ch.SetElement msoElementDataLabelOutSideEnd
            Set s = ch.SeriesCollection(1)
            s.Format.Line.Visible = msoTrue
            s.Format.Line.ForeColor.RGB = RGB(127, 127, 127)
            If s.Points.Count >= 1 Then s.Points(1).Format.Fill.ForeColor.RGB =
            If s.Points.Count >= 2 Then s.Points(2).Format.Fill.ForeColor.RGB =
            If s.Points.Count >= 3 Then s.Points(3).Format.Fill.ForeColor.RGB =
            If s.Points.Count >= 4 Then s.Points(4).Format.Fill.ForeColor.RGB =
            If s.Points.Count >= 5 Then s.Points(5).Format.Fill.ForeColor.RGB =
        Case "Product"
            ch.ChartType = xlBarClustered
            ch.ChartTitle.Text = "Revenue by Product/Service"
            ch.HasLegend = False
            ch.SetElement msoElementDataLabelOutSideEnd
            ch.SeriesCollection(1).Format.Fill.ForeColor.RGB = RGB(20, 181, 174
            ch.SeriesCollection(1).Format.Line.Visible = msoTrue
            ch.SeriesCollection(1).Format.Line.ForeColor.RGB = RGB(127, 127, 12
        Case "Conversion"
            ch.ChartType = xlDoughnut
            ch.ChartTitle.Text = "Conversion Breakdown"
            ch.HasLegend = True
            ch.Legend.Position = xlLegendPositionBottom
            Set s = ch.SeriesCollection(1)
            s.DoughnutHoleSize = 55
            s.ApplyDataLabels
            s.DataLabels.ShowPercentage = True
            s.DataLabels.ShowValue = False
            For i = 1 To s.Points.Count
                pName = LCase$(CStr(s.XValues(i)))
                If InStr(1, pName, "not", vbTextCompare) > 0 Then
                    s.Points(i).Format.Fill.ForeColor.RGB = RGB(255, 192, 0)
                ElseIf InStr(1, pName, "converted", vbTextCompare) > 0 Then
                    s.Points(i).Format.Fill.ForeColor.RGB = RGB(0, 176, 240)
                End If
            Next i
    End Select
```